

Islamic University of Technology

Course no: Phy 4143
Course Title: Physics II
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Nodal Analysis

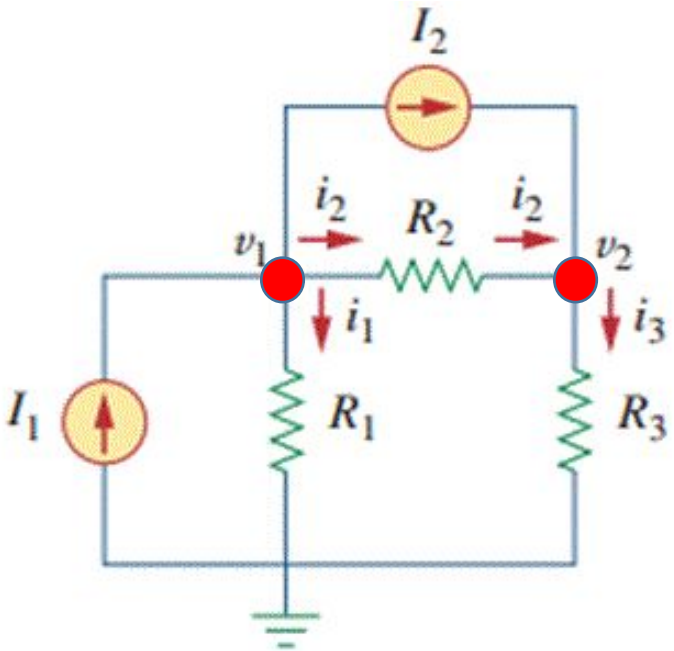
- Kirchhoff's Current law
- Ohm’s law

Nodal Analysis Definition: Nodal analysis is a method for analyzing electric circuits by calculating voltages at different points, or nodes, in the circuit.

Kirchhoff’s Current Law (KCL): KCL is used in nodal analysis to ensure the total current entering a node equals the total current leaving, which is fundamental for setting up the equations.

Reference and Non-Reference Nodes: Reference nodes are used as a baseline or zero voltage point, while non-reference nodes are the points where voltage is measured and calculated.

Handling of Components: In nodal analysis, components like resistors and current sources are used to formulate equations based on Ohm’s Law.

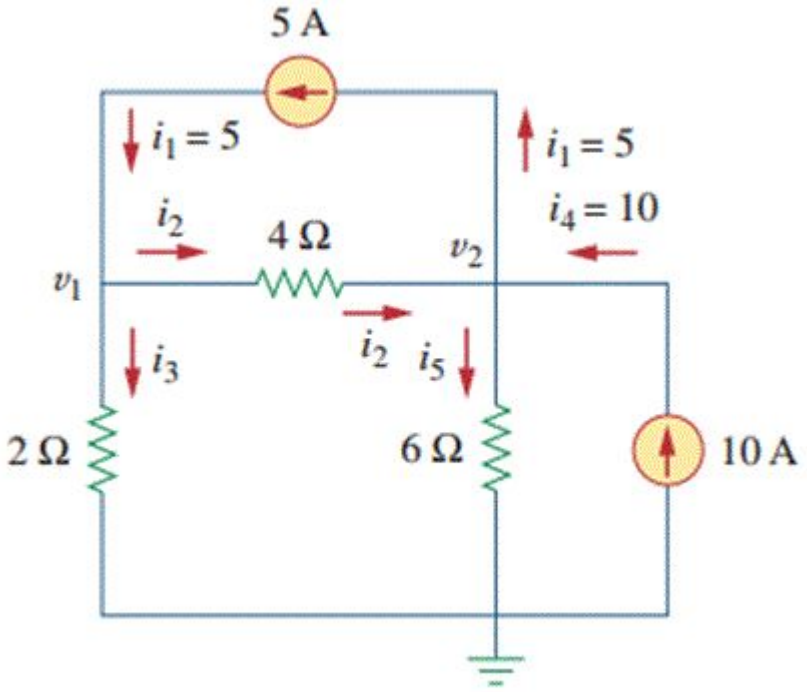


- **Identify the nodes**
- **Apply KCL at each node**
- **Write Equations for each node using Ohm’s law**
- **Solve the equation to find Node voltages**

Nodal Analysis with Current Sources

Nodal analysis with current sources is very easy and it is discussed with an example below.

Example: Calculate Node Voltages in following circuit



In circuit we have 3 nodes from which one is reference node and other two are non reference nodes – Node 1 and Node 2.

Step I. Assign the nodes voltages as v1 and 2 and also mark the directions of branch currents with respect to the reference nodes

Step II. Apply KCL to Nodes 1 and 2

KCL at Node 1

$$i_2 + i_4 = i_1 + i_5 \cdots\cdots (2) \quad \Rightarrow \quad i_1 = i_2 + i_3 \cdots\cdots (1)$$

Step III. Apply Ohm's Law to KCL equations Node 1 & 2 and simplifying

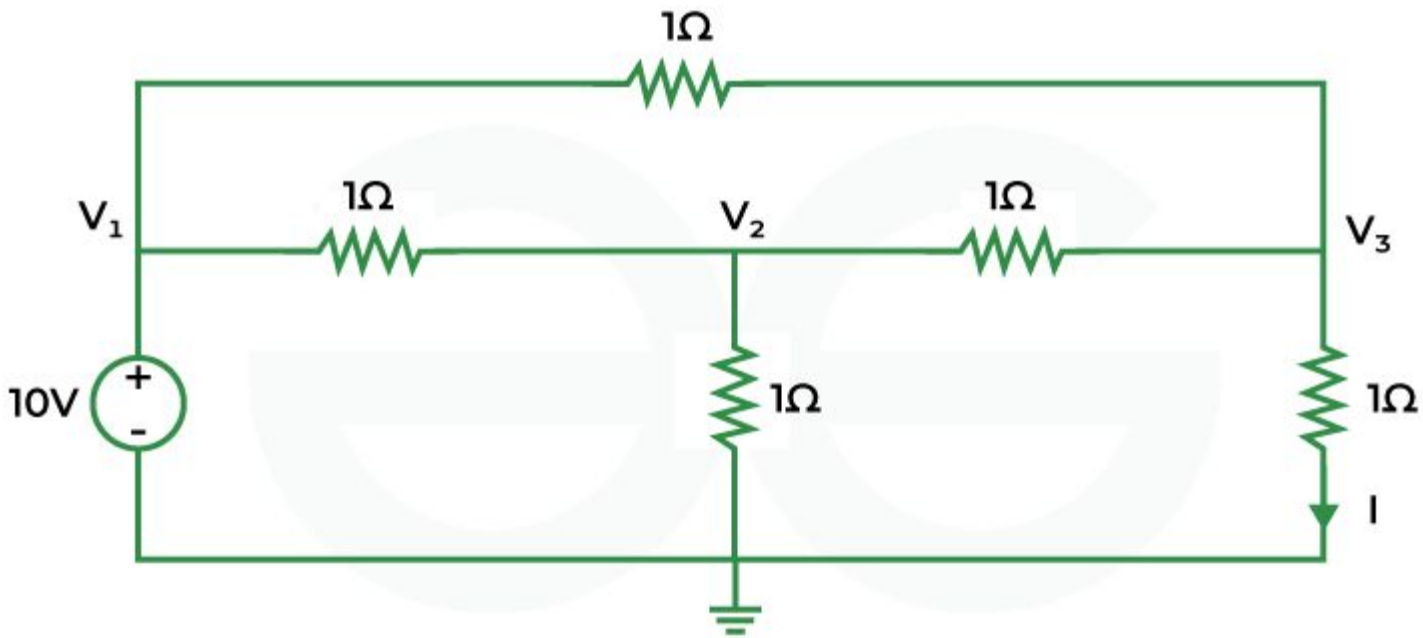
$$i_1 = i_2 + i_3 \Rightarrow 5 = \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2} \quad \Rightarrow \quad 3v_1 - v_2 = 20 \cdots\cdots (3)$$

$$i_2 + i_4 = i_1 + i_5 \Rightarrow \frac{v_1 - v_2}{4} + 10 = 5 + \frac{v_2 - 0}{6} \quad \Rightarrow \quad -3v_1 + 5v_2 = 60 \cdots\cdots (4)$$

Step IV. Solving (3) and (4) gives

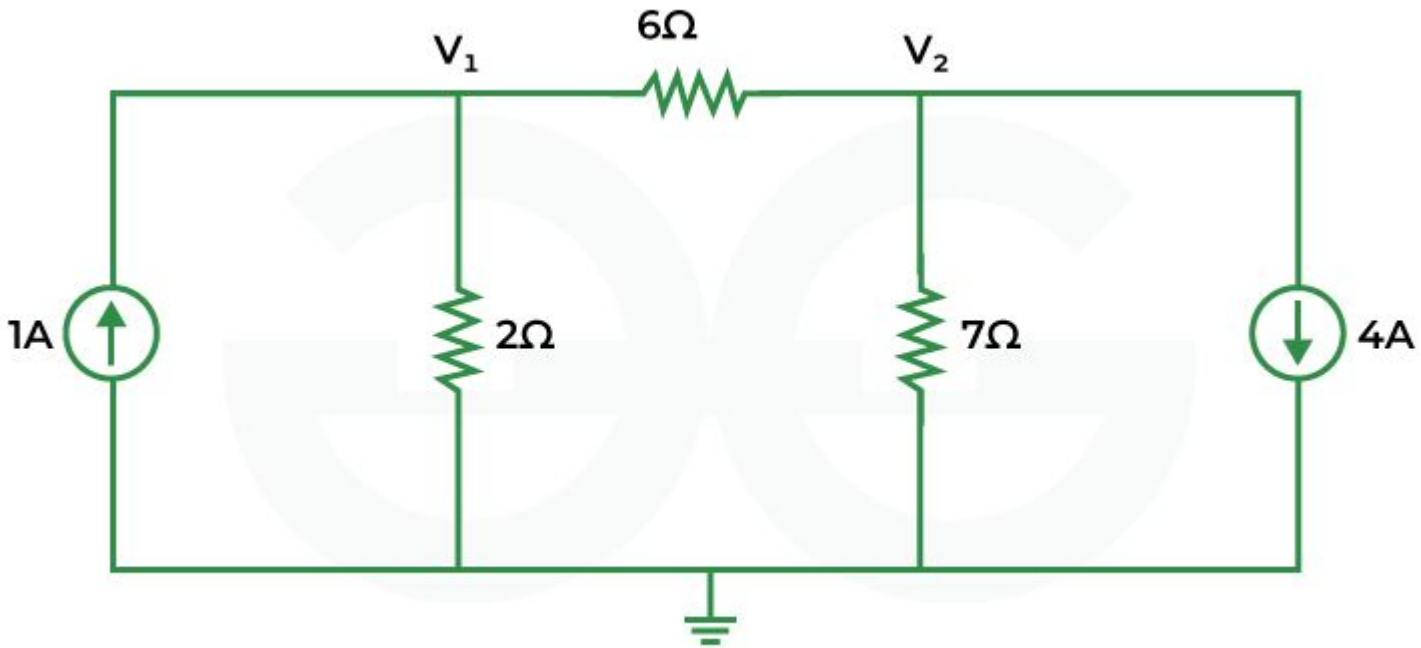
V_1 = 13.33 Volts and V_2 = 20 Volts

Q. Using nodal analysis find I(With Voltage source)



Ans. 5 A

Question .Find V1 and V2 using nodal analysis (With current Source)



Ans. V1 = -2 V and V2 = -14 V

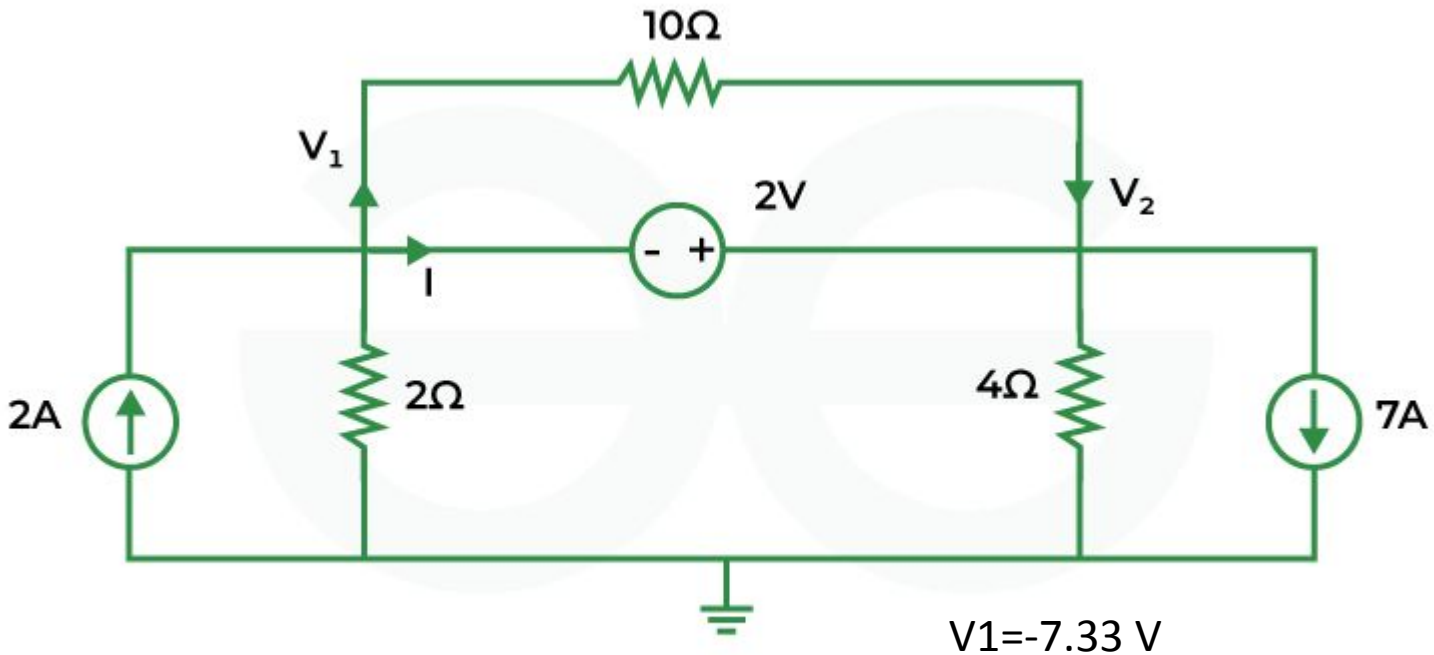
Super Node Analysis

Super node is special condition of nodes. When a voltage source is present between two non-reference node then it is called as super node. KVL and KCL are both applied to the super node to solve the equations.

Properties of Super node

It does not have well defined voltage
Both KCL and KVL are required to solve the equations
The voltage difference is zero between any two nodes in super node

Q. Using nodal analysis find V_1 and V_2



V1=-7.33 V
And V2= -5.33 V

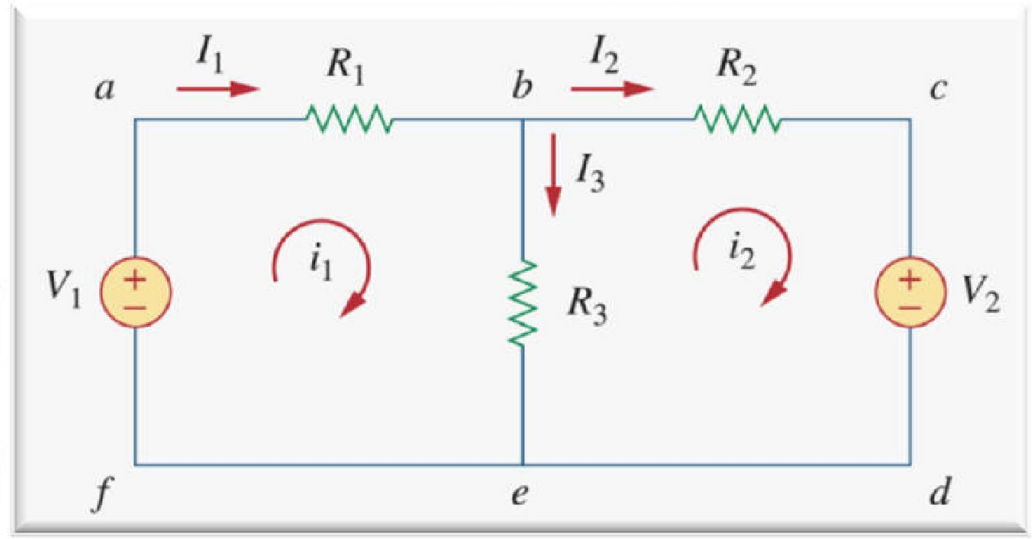
Mesh Analysis

- Kirchhoff's Voltage law
- Ohm’s law

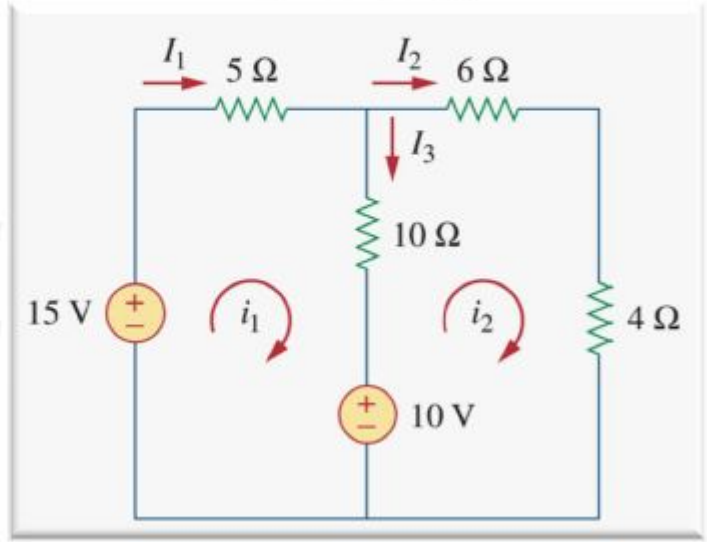
- **Mesh analysis is also known as loop analysis or the mesh current method.**
- **Nodal analysis applies KCL to find unknown voltages in a given circuit, while mesh analysis applies KVL to find unknown currents.**
- **The current through a mesh is known as mesh current.**

Steps

1. **Make a clear diagram.**
2. **Assign mesh currents i_1, i_2, i_3, …, i_n to the n meshes.**
3. **Apply KVL to each of the n meshes. Use Ohm’s law to express the voltages in terms of the mesh currents.**
4. **Solve the resulting n simultaneous equations to get the mesh currents.**



Example 1: For the shown circuit, find the branch currents and using mesh analysis.



We first obtain the mesh currents using KVL. For mesh 1,

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

$$3i_1 - 2i_2 = 1 \qquad (1)$$

For mesh 2

$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

$$i_1 - 2i_2 = -1 \qquad (2)$$

Solve equations 1 and 2 to get i_1 and i_2.

$$3i_1 - 2i_2 = 1 \qquad (1)$$

$$i_1 - 2i_2 = -1 \qquad (2)$$

$$2i_1 = 2$$

$$i_1 = 1\text{A}$$

Substitute i_1 in eq.1 or eq.2 to get i_2

$$3 \times 1 - 2i_2 = 1$$

$$-2i_2 = 1 - 3 = -2$$

$$i_2 = 1\text{A}$$

$$I_1 = i_1 = 1\text{A} \qquad I_2 = i_2 = 1\text{A}$$

$$I_1 = i_1 - i_2 = 1 - 1 = 0\text{A}$$

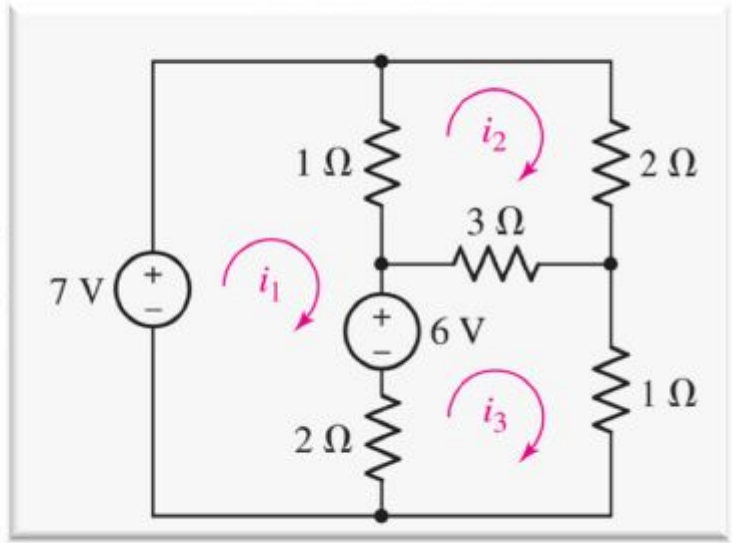
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Example 2: For the shown circuit, find the mesh currents using mesh analysis.

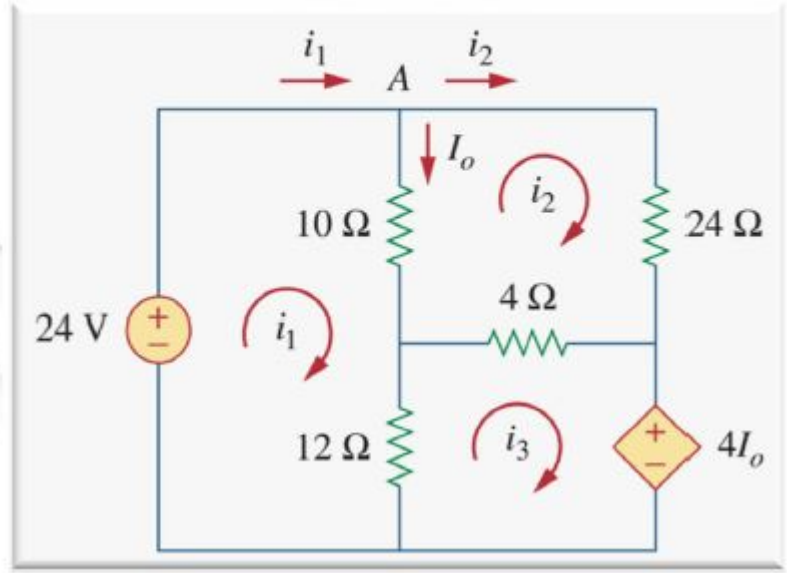
$$-7 + 1(i_1 - i_2) + 6 + 2(i_1 - i_3) = 0$$
$$1(i_2 - i_1) + 2i_2 + 3(i_2 - i_3) = 0$$
$$2(i_3 - i_1) - 6 + 3(i_3 - i_2) + 1i_3 = 0$$

Simplifying,

$$3i_1 - i_2 - 2i_3 = 1 \quad (1)$$
$$-i_1 + 6i_2 - 3i_3 = 0 \quad (2)$$
$$-2i_1 - 3i_2 + 6i_3 = 6 \quad (3)$$



Example 3: Use mesh analysis to find the current in the circuit of the shown figure.



We first obtain the mesh currents using KVL.
For mesh 1,

$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0$$

$$11i_1 - 5i_2 - 6i_3 = 12 \qquad (1)$$

For mesh 2

$$24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0$$

$$-5i_1 + 19i_2 - 2i_3 = 0 \qquad (2)$$

For mesh 3

$$4I_o + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

but $I_o = (i_1 - i_2)$

$$4(i_1 - i_2) + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

$$-i_1 - i_2 + 2i_3 = 0 \qquad (3)$$

Solve equations 1, 2 and 3 to get i_1 , i_2 and i_2.

Multiply eq. 2 by 3 and then subtract the resulting eq. from eq.1.

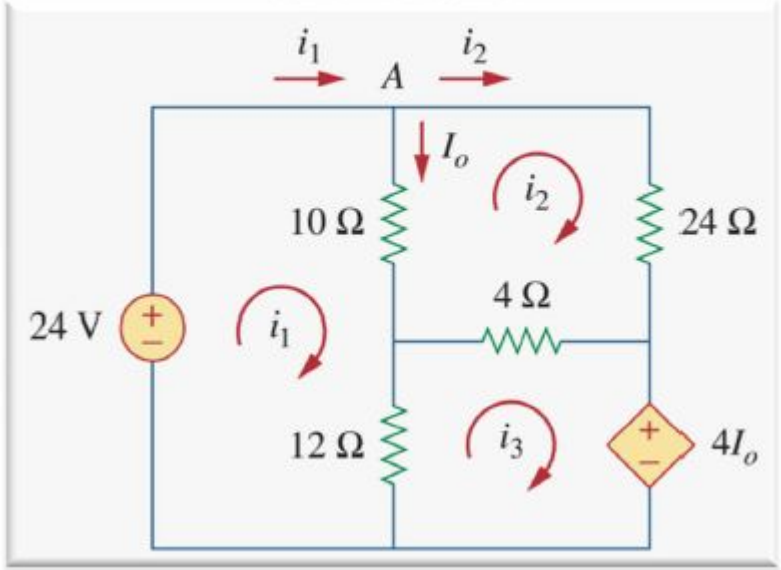
$$11i_1 - 5i_2 - 6i_3 = 12$$

$$-15i_1 + 57i_2 - 6i_3 = 0$$

$$26i_1 - 62i_2 = 12 \qquad (4)$$

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Example 3: Use mesh analysis to find the current in the circuit of the shown figure.



Multiply eq. 3 by 3 and then add the resulting eq. from eq.1.

$$11i_1 - 5i_2 - 6i_3 = 12$$
$$-3i_1 - 3i_2 + 6i_3 = 0$$

$$8i_1 - 8i_2 = 12 \qquad (5)$$

Multiply eq. 5 by 62/8 and then subtract the resulting eq. from eq.4.

$$26i_1 - 62i_2 = 12 \qquad (4)$$
$$62i_1 - 62i_2 = 93$$

$$-36i_1 = -81$$

$$\boxed{i_1 = 2.25\text{A}}$$

Substitute i_1 in eq.4 or eq.5 to get i_2

$$8\times2.25 - 8i_2 = 12$$

$$\boxed{i_2 = \frac{8\times2.25-12}{8} = 0.75\text{A}}$$

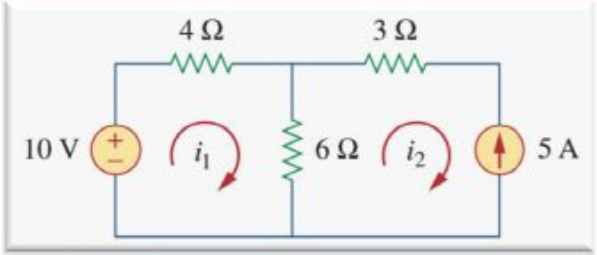
$$\boxed{I_o = i_1 - i_2 = 2.25 - 0.75 = 1.5A}$$

Mesh Analysis with Current Sources

CASE 1: When a current source exists only in one mesh.

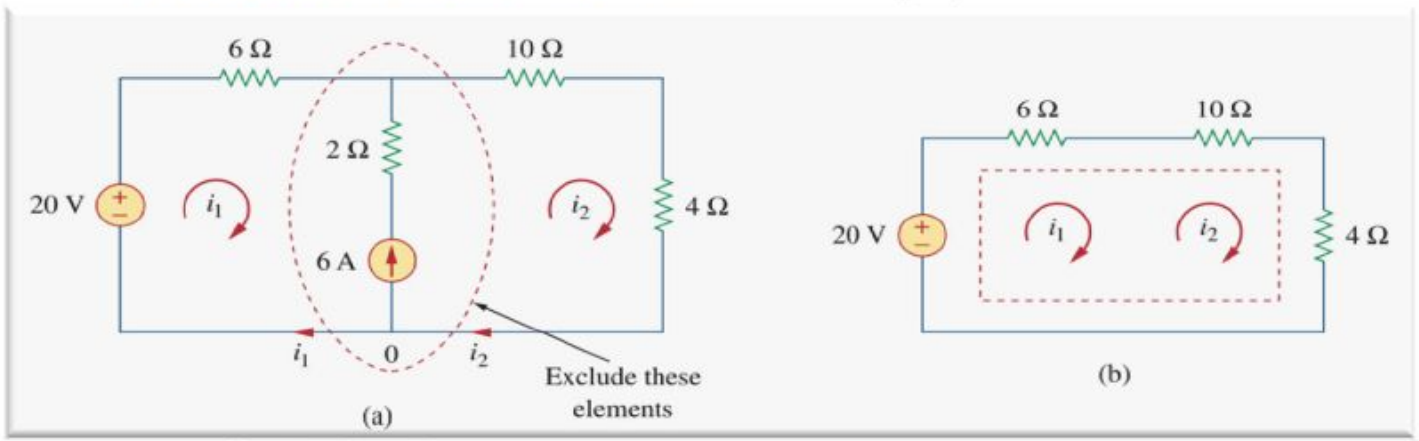
for example. i_2=-5A and write a mesh equation for the other mesh in the usual way; that is,

$$-10 + 4i_1 + 6(i_1 - i_2) = 0 \quad \Rightarrow \quad i_1 = -2\ \text{A}$$



CASE 2: When a current source exists between two meshes.

We create a *supermesh* by excluding the current source and any elements connected in series with it.



$$-20 + 6i_1 + 10i_2 + 4i_2 = 0$$

$$i_2 = i_1 + 6$$

A *supermesh* results when two meshes have a (dependent or independent) current source in common.

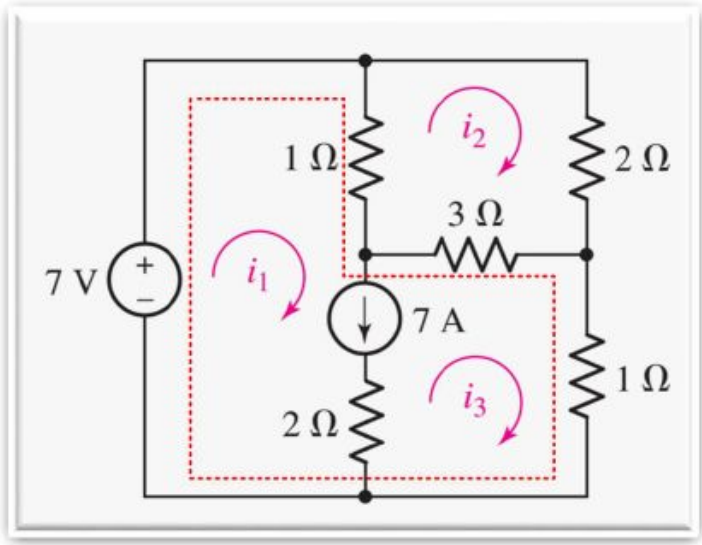
No, thanks

Mesh Analysis with Current Sources

Note the following properties of a supermesh:

1. The current source in the supermesh provides the constraint equation necessary to solve for the mesh currents.
2. A supermesh has no current of its own.
3. A supermesh requires the application of both KVL and KCL.

Example: Determine the three mesh currents in the shown figure.



7A independent current source is in the common boundary of two meshes, which leads us to create a supermesh whose interior is that of meshes 1 and 3. Applying KVL about this loop.

$-7+1(i_1 - i_2)+3(i_3 - i_2)+1i_3=0$

$i_1 - 4i_2+4i_3=7$ (1)

For mesh 2

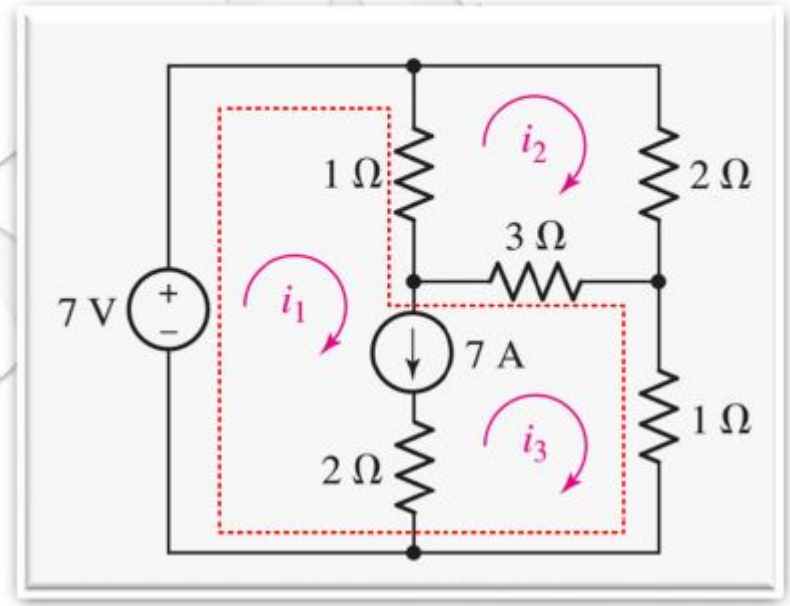
$$1(i_2 - i_1)+2i_2+3(i_2 - i_3)=0$$

$$-i_1 + 6i_2 - 3i_3=0 \qquad (2)$$

the independent source current is related to the mesh currents

$$i_1 - i_3=7 \qquad (3)$$

Solve these three equation to get mesh currents



i_1=9A

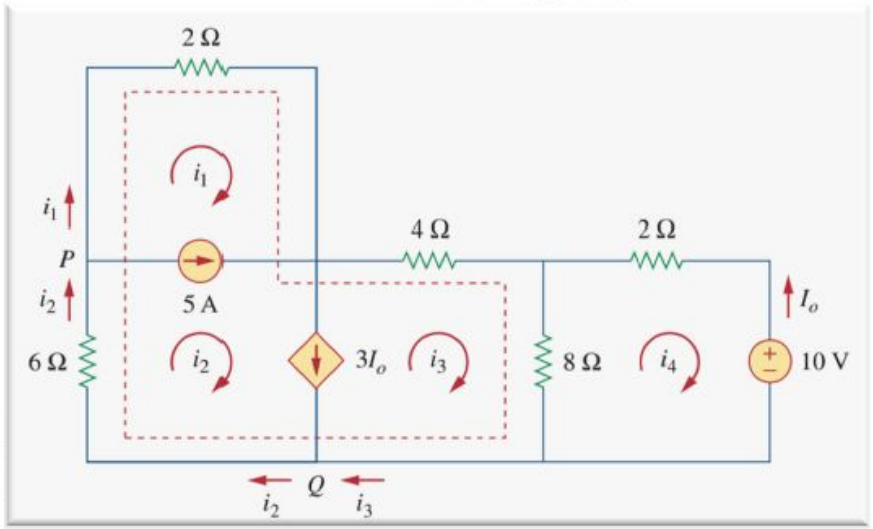
i_2=2.5A

i_3=2A

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Mesh Analysis with Current Sources

Example 4: For the shown circuit, find i_1 to i_4 using mesh analysis.



Note that meshes 1 and 2 form a supermesh since they have an independent current source in common. Also, meshes 2 and 3 form another supermesh because they have a dependent current source in common. The two supermeshes intersect and form a larger supermesh as shown. Applying KVL to the larger supermesh

$$2i_1+4i_3+8(i_3-i_4)+6i_2=0$$

$$2i_1+6i_2+12i_3-8i_4=0 \qquad (1)$$

For the independent current source, we apply KCL to node P:

$$i_2=i_1+5 \qquad (2)$$

For the dependent current source, we apply KCL to node Q:

$i_2 = i_3 + 3I_o$

$I_o = -i_4$

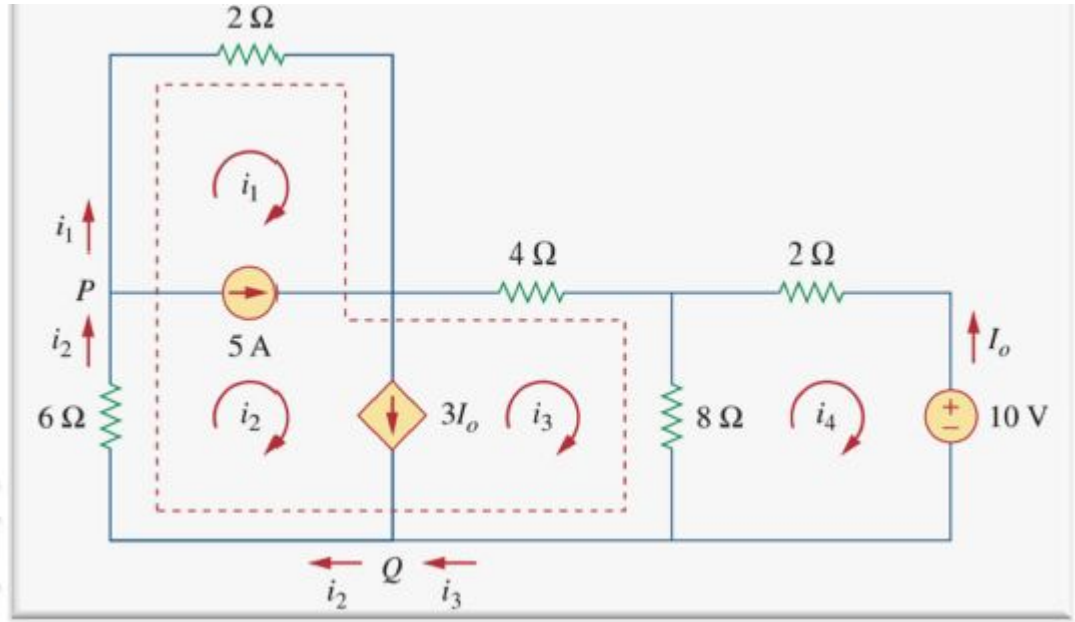
$i_2 = i_3 - 3i_4$ (3)

Applying KVL in mesh 4,

$2i_4 + 8(i_4 - i_3) + 10 = 0$

$5i_4 - 4i_3 = -5$ (4)

Solve these four equation to get mesh currents



i_1=-7.5A i_2=-2.5A

i_3=3.93A i_4=2.143A